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Final Project Report

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A Comparison of Observational Learning Model and Bayesian Brain Model in Cognitively Modeling Symptom Perception

The American Medical Association's Code of Medical Ethics defines a placebo as, "a substance provided to a patient that the physician believes has no specific pharmacological effect on the condition being treated". Thus, the term "placebo effect" that will be referenced in this paper refers to a patient feeling relief after being administered fake treatment. The AMA's Code of Medical Ethics, and studies from Harvard Health and Michigan Medicine, among many other universities and journal studies, recognize that the placebo effect is real and that placebo treatment can be effective in treating patients. This phenomenon elucidates an interesting cooperation between the mind and body and identifies a complication to the biomedical understanding of the cognitive process of symptom perception.

The biomedical model of symptom perception claims that the brain receives sensory signals from the body and, "converts them, directly, into conscious [experience](#)". Essentially, our perception of pain or relief matches the physiological state of our body. While this bottom-up driven process is a part of our perceptive process, it fails to account for the relief people feel when administered a placebo drug (the placebo effect), and the phenomena of feeling symptoms when no pathophysiological disruption has occurred (the nocebo effect). The lack of understanding of these two phenomena under the biomedical model has interested researchers in

studying how our brain tricks us into feeling these two effects. Aiming to correct the shortcomings of the biomedical model, we are introduced to the observational learning and Bayesian brain models of symptom perception, both of which treat symptom perception as a multifaceted cognitive process that integrates observation, visual perception, and top-down processing.

Babel and Bajcar's observational learning model emphasizes the impact of expectation, patient characteristics, and social factors on experiencing the placebo or nocebo effect. This framework suggests that observing another person's symptoms, especially pain, along with an observer's empathetic tendencies, shapes their expectations, leading to a divergence between physiological state and symptom perception. The model/framework proposed in the document integrates Banduras' social learning theory and multiple existing researched representations and processes related to observational learning.

In the observational learning process, as presented by Banduras' social learning theory, the three main ways of transmitting information (in this study the information is pain symptoms) are through behavioral modeling, verbal modeling, and symbolic modeling. These representations involve the physical demonstration of pain behavior, verbal description of pain, and indirect pictorial representation, respectively. The cited Colloca and Benedetti study demonstrated how behavior signaling physical pain could produce a placebo response, although not a nocebo response, to individuals who were witnessing the pain experience of another. Furthermore, this study later found a positive correlation between a placebo response and an empathetic response in this behavioral modeling scenario. Although they concluded that empathy is not a crucial factor in the production of the placebo effect, it does introduce the role of top-down processing involving individual characteristics of the observer in the formation of

observationally induced analgesia and hyperalgesia. Additionally, studies by Yoshida and Koban & Wager showed how verbal modeling could induce the placebo effect in individuals through the intake of social information and cues in the form of pain ratings accessible to the individual experiencing the placebo effect. Finally, Vögtle showed that the nocebo effect was exhibited even when a live model was not available for comparison - a symbolic representation can produce the same effect. Curiously, this model was not computational in nature and instead used a variety of resources to construct its framework; the model is tested and evaluated on the studies it was created by, making us question the success of this model which claims to explain the cognitive processes underlying the placebo and nocebo effects.

Overall, Babel and Bajcar's paper proposed the observational learning model, a persuasive framework that demonstrated how the placebo and nocebo effects could be explained by three facets of the social learning theory; the evidence for the placebo effect being induced by both behavioral and symbolic modeling demonstrates that direct experience and observation are not necessary to produce a pain response, and verbal modeling studies further highlighted the role of expectancies, leading us to understand the placebo and nocebo effect as being induced in a more top-down cognitive process of pain, rather than just bottom-up sensory signal conversion.

The second paper cognitively modeling the placebo effect is Ongaro and Kaptchuck's Bayesian predictive processing model of symptom perception. This paper specifically accounts for the two aforementioned shortcomings of the biomedical model through a Bayesian approach to perception. This predictive processing approach involves the integration of sensory input, prior experience, and contextual cues in the updating of Bayesian priors to model a top-down predictive cognitive process informed by bottom-up signals. While this paper discusses symptom perception more broadly, the authors claim that the placebo and nocebo effects are examples of

the power of prediction, expectation, and inference of impending health in the Bayesian brain.

As we've learned this quarter, the Bayesian approach updates the likelihood or probability of an event (posteriors) based on prior knowledge of salient information (priors).

Neurally, Ongaro and Kaptchuck explain that this happens by the top-down generation of neurally encoded hypotheses about the body, the healthy body condition, which are then updated when met by the bottom-up stream of sensory input that the body is continuously receiving. This healthy body condition is akin to the homeostatic condition of our body - we have some physiological constants that are seen as "noise" to the brain, like the regular frequency of our heartbeat. These healthy body conditions are the priors in the Bayesian brain. When there are variations to our physiological constants, a mismatch between the predicted and expected healthy body condition hypothesis and the actual sensory input to our body, prediction error occurs which prompts revision of our healthy body condition hypothesis. So, applied to symptom perception, "we feel symptoms, including pain, when the hypothesis with the lowest prediction error represents an abnormal somatic event". Essentially when our expectation matches the actual sensory input, prediction error is low and our prior hypothesis is somewhat confirmed, allowing us to perceive the symptom.

Both the placebo and nocebo effects can be accounted for under this process of perception. To start with a simple example, imagine you come home from a run only to discover that you have a bleeding scratch on your ankle. You don't know the exact moment you got it, nor did you feel any pain when it happened. On your run, you likely were not expecting to feel pain besides the usual running pains like muscle aches or heavy breathing. Because the bleeding and scratch were not part of your expected healthy body condition for the state of your body running, you didn't feel the pain (there was likely a high mismatch between expected and actual sensory

input to your body. On the other hand, once you see the blood on the scratch, the common association of blood with pain causes you to update your healthy body condition and expectations to expect soreness in that area. You accidentally bump your scratch against a table leg later that day after you've noticed the cut and you feel the soreness of the cut being pressed because of relatively low variation in your expected body condition and the sensory input. While this example may seem trivial, both the nocebo and placebo effects are arguably being demonstrated here. You don't feel pain from the initial scratch because you expect to feel healthy on the run – placebo – and even though you felt no pain when you got the cut, because you are expecting to feel a cut's soreness, pressure to the cut causes pain – nocebo.

More directly explaining the placebo effect, Ongaro and Kaptchuck tell us that relief is experienced as a consequence of interpreting that we are healing – our brains essentially infer and update our expectations so that our predictions will later be confirmed and prediction error will be minimized. Similarly, if, when in pain, we aren't given reason to update our expectations, then, without an external signal to spark the update of a prior, we will continue to suffer pain regardless of our pathophysiological status. For example, in an experiment discussed in the paper, researchers found that patients who were secretly administered symptom-relieving drugs felt less relief from symptoms than patients who were administered the same drugs more overtly. While you'd expect the same drug to cause the same relief in patients with the same illness, the patients who expected healing from the drug interpreted even small changes in their body to be signs of healing, creating a cycle of expectation to the affirmation of a hypothesis representing a healthier body (“fulfilling prediction and minimizing error”).

Now understanding both the observational learning and Bayesian brain model, we argue that the Bayesian model serves as a better cognitive model of the placebo and nocebo effects.

While the observational learning model paper seems specific to studying the placebo and nocebo effect, I find it more convincing that these phenomena are side effects of our perceptive process rather than specific separate cognitive processes. One shortcoming of the observational learning model paper is that the studies it cites, like the Colloca and Bendetti study, seem to be unable to provide support for the existence of the nocebo effect when placebo effects are displayed. Given that these are both peculiarities of our symptom perception system, the Bayesian brain model's more general approach to symptom perception is a more convincing model than the separate research findings bound together by the social learning theory.

Furthermore, Ongaroa and Kaptchuck admit that social learning theory is a major contributor to the formation of placebo and nocebo effects without undermining their argument. In other words, the observational learning model seems to be reducible to the Bayesian model. Specifically, social and environmental learning changes one's expectations and predictions almost as much as visual perception or sensory events do; certain experiments have even shown that suggestions by a trusted clinician or the reported or marketed value of a medication and its efficacy can induce placebo effects. This demonstrates that the social value of treatment or behavior and observation of relief or pain can be triggered to update one's healthy body hypothesis. Seeing social learning theory as a trigger for Bayesian updating rather than the cause of the placebo effect allows us to account for mismatching in perception more broadly. For example, in our visual perception system, the color constancy phenomenon has been widely studied and called a societally shared feature of the human perceptive system. While it may be possible for a reader to call the social learning model predictive, the authors cite this as an area of further research and therefore outside the scope of their proposed model: "Therefore, the role

of expectancies in the formation of an observationally induced placebo effect requires further investigation”.

A shared shortcoming in these two papers is the lack of evaluation and testing of the models they present. Given that we’ve seen the testing of Bayesian models throughout this class and that the Bayesian model has an arguably clear Python implementation, the Bayesian brain model seems more feasible to evaluate. Still, collecting data on physiological characteristics and encoding the “healthy body condition” of individuals into priors seems like an unclear and intensive clinical process. That being said, evaluating the observational learning model seems an even more unclear process. While multiple studies could provide as qualitative evidence of the observational learning model, in a similar way to the studies that helped construct the framework presented above, there does not seem to be a quantitative way to assess this model. Since, arguably, the observational learning model is reducible to the Bayesian predictive processing model, further work to legitimize the model could be instrumental in assessing why our perceptions deviate from the actual state of the world or body. This Bayesian perceptive process applies to more than symptom perception and has the potential to explain visual, auditory, olfactory, and attentional perceptive processes.

More broadly, these cognitive models elucidate the bigger idea in cognitive science that we don't perceive the world as it is, but as the brain's best guess of it in any given context. Our perceptual processes are extremely complex and although not perfect, the Bayesian model provides a more holistic account of the aforementioned failures of the biomedical model. To summarize, here is ChatGPT’s summary of my discussion thus far:

In conclusion, both the observational learning and Bayesian predictive processing models offer valuable insights into the placebo and nocebo effects, expanding our understanding beyond the traditional biomedical model. The observational learning model emphasizes the role of social and individual factors in shaping symptom perception, while the Bayesian model integrates sensory input, prior experiences, and contextual cues to explain how expectations and predictions influence our perception of symptoms. The Bayesian approach, with its broader applicability and potential for quantitative evaluation, appears to provide a more comprehensive framework. It accounts for the placebo and nocebo effects as inherent aspects of our perceptual processes, rather than isolated phenomena. Despite the promising perspectives of both models, further empirical testing and refinement are necessary to fully validate and operationalize these theories in clinical settings.

AI Usage Statement:

The only part of this report which utilized AI tools was the italicized conclusion of my written paper up to that point. I believe this conclusion serves as more of a summary rather than any new analysis and simply reiterates the representations and comparisons of the two models!

Works Cited

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